Where Does Vision Belong in a Frozen Time-Series Forecaster?

Abstract. Multimodal time-series forecasters are normally trained end to end, which makes it impossible to attribute an accuracy gain to the additional modality rather than to the extra capacity or the extra gradient path it brings. This report removes that confound by freezing both backbones — a pretrained time-series transformer and a self-supervised video encoder — and varying only the point at which the visual representation is attached. Four configurations are trained on an identical schedule from an identical initialisation and evaluated on a cross-plant photovoltaic benchmark of held-out installations, using a counterfactual pass that switches the imagery off at inference to measure how much of each model's accuracy actually depends on having seen the sky. Pooled fusion on the batch axis and appended fusion on the sequence axis both yield reliance indistinguishable from zero, and widening the visual channel from one token to sixteen does not change that. Reliance becomes measurable only when the imagery is kept unpooled as an external memory and queried by the forecast positions themselves, one query per lead-time slot. The placement, not the bandwidth, is what determines whether a frozen forecaster uses vision at all.

1 Introduction

The multimodal forecasting literature is largely a literature of proposals: an architecture is introduced, trained end to end, shown to beat a set of unimodal baselines, and the improvement is attributed to the additional modality. That attribution is rarely tested. When both encoders are trained jointly, an accuracy gain is consistent with at least three explanations — the model learned to read the images, the extra capacity helped, or the extra gradient path acted as regularisation.

This work fixes the confound by construction. Both encoders are **pretrained and frozen**, so neither can adapt to the other; only a small bridge between them is learned. The architecture is then held fixed and the **attachment point** is varied. The question becomes:

> *Given a frozen sequence forecaster and a frozen visual encoder, at which point in the forecaster's computation must the visual representation be introduced before the forecaster demonstrably uses it?*

Freezing is what makes the question answerable. With both backbones fixed, no configuration can win by having more capacity than another, and any measured difference is attributable to the wiring. It is also what makes the result portable: the finding is a statement about placement in a frozen pipeline, not about one particular set of weights.

The contributions are as follows.

1. A **placement study** rather than an architecture proposal: three fusion sites are compared under a frozen-backbone protocol that holds capacity, initialisation and training schedule constant (Section 5.2, Table 6).
2. A **counterfactual reliance measure** that quantifies how much of a trained model's accuracy depends on the imagery, by re-evaluating the same frozen weights with the visual pathway switched off (Section 6).
3. A **consolidated cross-region multimodal photovoltaic corpus** — 110 installations on two continents, satellite frames co-registered per site and stored with timestamp-exact pointers, infrared bands that remain informative at night, and a cross-plant generalisation split (Section 3.2, Table 2).
4. A **full-suite comparison** against 32 baselines spanning statistical references, classical machine learning, supervised deep forecasters, time-series foundation models, retrieval-augmented adaptation and published multimodal forecasters (Table 9 to Table 11).

2 Related work

Published multimodal forecasters differ in many respects, but for the purposes of this study they differ in exactly one: **where the auxiliary representation is attached relative to the point at which the horizon is**

resolved. Five representative systems are reproduced schematically below in a common visual language so that the attachment point can be compared directly: three satellite-based multimodal forecasters (Figure 1 to Figure 3), and the two strongest non-MMTSFM entries of the baseline suite (Figure 4, Figure 5), which are strongest for reasons that bear directly on the placement question. All five are run as baselines under this report's protocol (Table 9, Table 10).

2.1 Convolutional joint encoding — SUNSET

Figure 1: SUNSET [1]. The image branch is reduced to a flat feature vector and concatenated with the numeric history before the prediction head; both branches are trained jointly.

SUNSET is the canonical convolutional solar baseline and the ancestor of most sky-image forecasters. Its fusion is **concatenative**: a flattened convolutional descriptor is joined to the power history, and a multilayer perceptron maps the concatenation to the whole horizon. The visual descriptor is therefore computed once, independently of the horizon, and every lead time reads the same vector.

2.2 Cross-attention from history timesteps — CrossViViT

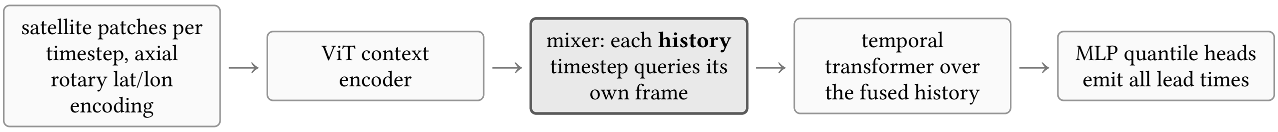

Figure 2: CrossViViT [2]. Cross-attention is present, but the queries originate in the **history** representation; the horizon is decoded afterwards from the already-fused sequence.

CrossViViT is the strongest published satellite-plus-time-series architecture and the closest structural relative of the configuration proposed here. It encodes each satellite frame with a vision transformer using axial rotary embeddings over latitude and longitude, encodes the station series with a separate transformer, and mixes the two with cross-attention. The decisive detail is the direction of that cross-attention: the **query** is the station token at a given history timestep and the keys and values are that timestep's image patches. Fusion therefore completes before the temporal decoder runs, and the horizon is produced from a sequence in which the visual evidence has already been condensed.

2.3 Gated late fusion of pooled embeddings — Solar-VLM

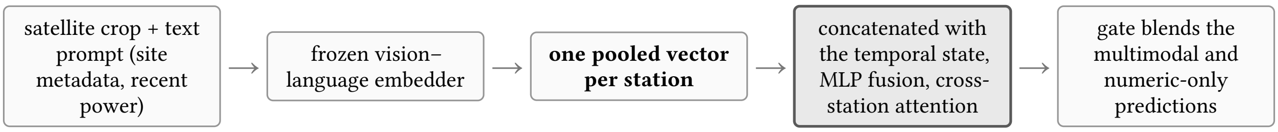

Figure 3: Solar-VLM [3]. The scene is compressed to a single per-station embedding; a learned gate decides how far the multimodal prediction is allowed to move the numeric one.

Solar-VLM represents the vision–language family [4], [5]. Its visual pathway ends in one pooled vector per station, which is concatenated with the temporal state and passed through a fusion network; a modality gate then interpolates between the multimodal prediction and the numeric-only prediction. As in the previous two cases, the visual summary is fixed before the model has committed to any particular lead time.

2.4 Vision without a scene — Time-VLM

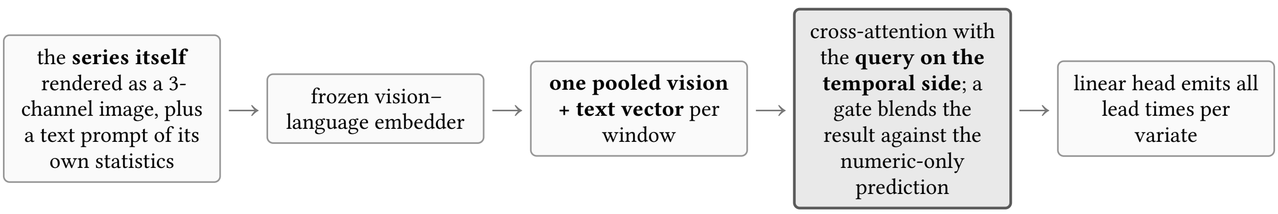

Figure 4: Time-VLM [6]. The imagery is manufactured from the numeric input rather than observed, and the pooled multimodal vector serves as keys and values for a query formed from the temporal state.

Time-VLM is the strongest published multimodal entry in the suite (rank 2, SS 0.5404, Table 9) and it is instructive precisely because its second modality carries no external information. There is no camera and no satellite: the past window is plotted into a three-channel image, a prompt is assembled from the window's own minimum, maximum, median and trend direction, and both are embedded by a frozen vision–language model. Whatever the visual branch contributes is therefore a re-encoding of data the numeric branch already has. Its accuracy comes from reusing visual pretraining as an inductive bias over the series — the same direction as VisionTS++ [7] — not from seeing anything new.

Structurally it repeats the pattern of the preceding three. Both modalities are pooled to a single vector per window; that vector is used as keys and values while the query is formed from the temporal features; and a learned gate interpolates between the multimodal path and a numeric-only prediction. The visual representation is again complete before any lead time is distinguished.

The consequence for this report is a measurement one. A model that scores well with a visual channel that provably carries no new information is a warning about attributing a leaderboard gain to a modality, and it is the reason the counterfactual instrument of Section 6 is defined on the imagery rather than on the architecture.

2.5 Exogenous information aligned to the horizon — iTransformer

Figure 5: iTransformer [8] with covariates. Inverting the transformer's axes makes each variate a token; with the covariate window shifted forward, the exogenous channel is already aligned to the lead times being predicted.

iTransformer is not a multimodal architecture and is included for a different reason: it holds the best ramp NMAE of the whole suite (0.1445, Table 10), better than every multimodal model tested including S2c. It inverts the usual axes — a token is a whole variate rather than a timestep — so attention runs across the target and its eleven covariates while the feed-forward path runs along time. Under this report's protocol it is run with the covariate window shifted onto the forecast interval, which is the deployable-NWP assumption the MMTSFM arms are also trained under and what makes the two modality-identical.

That shift is the point. The exogenous channel that wins on ramps is one whose content is **already indexed by the lead time it is meant to inform**: the token describing cloud cover describes the cloud cover of the hours being predicted, not of the hours already observed. No architecture is needed to route it to the right horizon, because the alignment is in the data. Satellite imagery has the opposite property — it is an observation of the past, and something in the model must decide which part of it pertains to which future step. iTransformer therefore sets the ramp-metric reference point this report is measured against, and simultaneously illustrates, from the numeric side, the property that S2c has to construct architecturally.

2.6 The shared assumption

Figure 1, Figure 2, Figure 3 and Figure 4 differ in encoder family, in training regime, in whether the second modality is even observed, and in whether cross-attention is used at all, yet they agree on one structural choice: **the auxiliary representation is finalised before the horizon is resolved, and every lead time receives the same representation**. Retrieval-augmented and covariate-adaptation methods for frozen backbones [9], [10], [11] make the same choice for a different auxiliary modality. Figure 5 is the exception that locates the cost of that choice: it is the only strong entry whose exogenous channel is aligned to the forecast interval rather than summarised ahead of it, and it is the one that wins on ramps. The experiments in this report are designed to test whether the shared choice is what prevents a frozen forecaster from using satellite imagery.

3 The dataset

3.1 Construction

The corpus fuses four public sources into one standardised table keyed by `(site_id, timestamp_utc)`, plus a single image archive addressed by a timestamp-exact per-row pointer (Table 1). Power is normalised by audited installed capacity, so the target is dimensionless and comparable across installations of very different size.

Track	Source and processing
PV power, UK	`openclimatefix/uk_pv` [12]: 30-minute generation for 2019–2020 with per-site capacity and rounded coordinates. Energy over the interval is converted to power; gaps of at most three steps are linearly interpolated, longer ones dropped.
Satellite, UK	EUMETSAT SEVIRI Rapid Scanning Service [13], reprojected per site and cropped to 128 × 128 px (~ 128 km). Version 2 of the corpus uses the **non-HRV** multi-band product in place of the earlier single-channel HRV crops.
PV power and satellite, US	NREL PVDAQ systems from the OEDI data lake [14] paired with GOES-16 crops at 256 × 256 px.
Weather covariates	Open-Meteo historical archive [15]: eight variables joined to each row by nearest timestamp. Solar geometry and a Haurwitz clear-sky GHI are computed analytically.

Table 1: Sources fused into the corpus. Every numeric row carries its own frame pointer, so the numeric and visual tracks are aligned by construction rather than by post-hoc matching.

3.2 What is new about it

Four properties distinguish this corpus from the public multimodal PV datasets it is closest to. They are summarised in Table 2; Table 3 places the corpus against those datasets directly.

Property	What it is	What it enables
Cross-region, single schema	110 installations on two continents — 100 UK residential rooftops at 1.5–4.0 kW and 10 US systems at 1.8–408 kW — in one table with one column contract, one normalisation and one set of quality flags.	Capacity- and climate-transfer questions can be asked without an ETL rewrite; the same model code consumes both regions.
Infrared bands, not HRV	Three genuinely distinct bands (inter-channel correlation 0.67–0.91, mean $\lvert R - G\rvert = 16.1$ DN) rather than a replicated grayscale channel, and coverage through the night: December pre-dawn frames carry mean 134 / std 37, against midday mean 123 / std 38.	Cloud fields are observable before sunrise, so a 14-day history window is visually covered rather than half blank. 96.1% of scored windows carry all eight frames.
Frames denser than power	Imagery at 15-minute cadence against a 30-minute power grid, 4,103,892 frames over 110 per-site groups (98 GB), addressed by a local-to-group index that is timestamp-exact for both regions.	Advection can be measured rather than assumed: frame-to-frame mean absolute difference is 8.3 DN at Δt = 15 min and 35.3 DN at 2 h, against a within-frame std of 38.6 DN — cloud-motion information is effectively exhausted by roughly two hours ahead.
Cross-plant split by design	Train, validation and test partitions are disjoint **by installation**, not by time; all three share the same two calendar years and share no rooftop.	The evaluation measures generalisation to an unseen installation, which is the deployment case, and removes the per-site memorisation that a chronological split leaves available.

Table 2: The four properties that are new, and what each one makes measurable. Each is a prerequisite for a claim this report makes.

Corpus	Sites / region	Imagery	Layout	Split
This work	110, UK + US	SEVIRI non-HRV 128^2, GOES-16 256^2, 15-min	tall table + frame pointer; any window	**cross-plant**, disjoint sites
ClimateHackAI 2023 [16]	Great Britain	SEVIRI HRV + 11-band non-HRV, 5-min	fixed 1 h in ⟶ 4 h out windows, plus NWP and air-quality tensors	competition split; no paper
MMSP / FusionSF [17]	88, one Chinese province	Himawari-8/9, 64^2, hourly	fixed day-ahead 24 h in ⟶ 24 h out	as released

Corpus	Sites / region	Imagery	Layout	Split
Sky-camera corpora [18]	single site each	ground fisheye RGB, sub-minute	continuous, single location	chronological, within site

Table 3: Positioning against the public multimodal PV corpora. The comparison is structural: each of the others is organised as fixed short windows for a fixed task, whereas this corpus is a tall `(site, timestamp)` table from which any window definition can be cut.

Neither ClimateHackAI nor MMSP is unusable for the question posed here, but both would require re-windowing and re-sampling before they could serve a long-history cross-plant protocol: they are stored as pre-cut input–output tensors for a fixed nowcasting or day-ahead task, whereas the protocol in Section 4 needs 14 days of history per origin and a split that holds whole installations out.

3.3 Quality control

Two UK installations carry a data-quality flag and are dropped, leaving 98. Row-level flags mark outages (15,486 rows), stuck sensors (1,318) and night-clamped values (1,535); flagged targets are excluded from scoring rather than imputed.

4 Experimental protocol

The experiments in this report are run on the UK track of Section 3. All metrics are normalised by installed capacity, so they are comparable across systems of different size, and all are computed on daytime steps only. Table 4 gives the full setting; every configuration in this report shares it.

Quantity / Metric	Specification / Description
Installations	100 UK residential rooftops, 1.5–4.0 kW; two carry a data-quality flag and are dropped, leaving 98
Sampling	every 30 minutes, years 2019 and 2020
History window	14 days = 672 samples of the plant's own power
Forecast horizon	6 hours = 12 samples, 9 quantiles emitted per step
Covariates	*Weather*: temperature, cloud cover, wind speed, precipitation, radiation, irradiance. *Solar geometry*: solar zenith, solar azimuth, day-of-year, solar time, clear-sky GHI
Visual signal	8 satellite frames at 224 × 224 covering the 6 hours ending at the origin, centred on the installation
Split	70/15/15 **by installation**, not by time: train, validation and test share the same two years and share no rooftop
Evaluation set	14 test installations, 165,295 scored daytime steps
Seeds	42, 43, 44
Skill score (SS, ↑)	$\mathrm{SS} = 1 - \mathrm{NRMSE}_{\mathrm{model}}/\mathrm{NRMSE}_{\mathrm{ref}}$ against Smart Persistence ($\mathrm{SS} = 0$). Higher is better; one means perfection.
Ramp NMAE (↓)	NMAE restricted to steps where true power changes sharply between consecutive samples — a cloud edge crossing the array (lower is better).

Table 4: The experimental setting and evaluated metrics. All configurations in this report share this setting.

Smart Persistence is the field's standard null hypothesis and it is a strong one: on clear or uniformly overcast days it is nearly unbeatable. Ramps deserve the same emphasis. They are a small minority of steps, but they carry nearly all of the operational cost of a forecast error, and they are the steps at which a satellite view could in principle help.

5 Method

5.1 Components

Table 5 lists the four components and their training state. Only the last two carry gradients; together they account for a small fraction of the parameters in the stack.

Component	Role and dimensions	State
Chronos-2 [19]	Pretrained encoder-only time-series transformer.	frozen, top 3 of 12 blocks unfrozen
V-JEPA 2 [20]	Self-supervised video encoder. Encodes the 8-frame clip once, offline, into 4 temporal slices × 196 spatial patches × 1024 channels, cached.	frozen entirely
Channel projection	Maps the visual channel width onto the backbone's width.	trained
Fusion module	The only thing that differs between configurations.	trained

Table 5: Components of the stack and their training state. The fusion module is the only element that differs between the four configurations of Section 5.2.

5.1.1 The forecasting backbone

The backbone treats a series the way a transformer treats any sequence: it is cut into non-overlapping patches of 16 consecutive samples, each patch is linearly projected into a token, and a 672-step history becomes 42 context tokens. Future positions are represented by additional placeholder tokens carrying only their known covariates; self-attention over the whole assembly lets those future positions absorb information from the past, and a projection head reads the answer off them.

The backbone is general-purpose: it was pretrained on heterogeneous series and has no notion of irradiance, panel tilt or clear-sky curves, so nothing about the solar domain is baked in. It also consumes covariates natively, which means the numeric-only control is already a strong model and the visual signal must earn its place against a high bar — as Table 10 confirms, the fine-tuned backbone alone outranks every published multimodal forecaster in the suite except one.

5.1.2 The visual encoder

The choice of a self-supervised video model rather than a caption-aligned one is deliberate. The relevant visual content — the motion and deformation of cloud fields — is temporal and has no useful textual description; a representation trained to match captions would discard exactly the structure that matters. V-JEPA's predictive objective in latent space is, by construction, a model of **how a scene changes**, which is the property the forecaster needs. The encoder is never run during training: clips are encoded once, offline, and cached.

5.2 The four configurations

The three fusion arms are best understood not as three architectures but as three answers to a single structural question: **along which axis does the visual information enter?** Figure 6 summarises the three answers and the outcome of each; Table 6 gives the corresponding numbers.

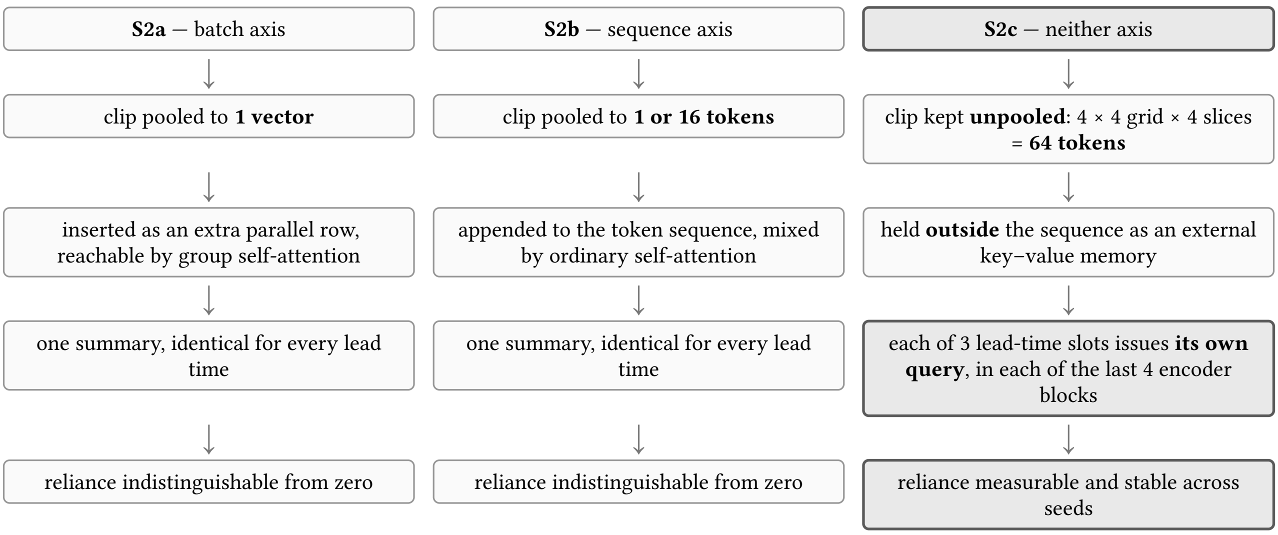

Figure 6: The placement taxonomy. The first two arms differ from each other in axis but agree on the choice identified in Section 2.6 — a single visual summary, fixed before the horizon is resolved. Exact reliance figures are in Table 6.

5.2.1 S1 — the vision-free control

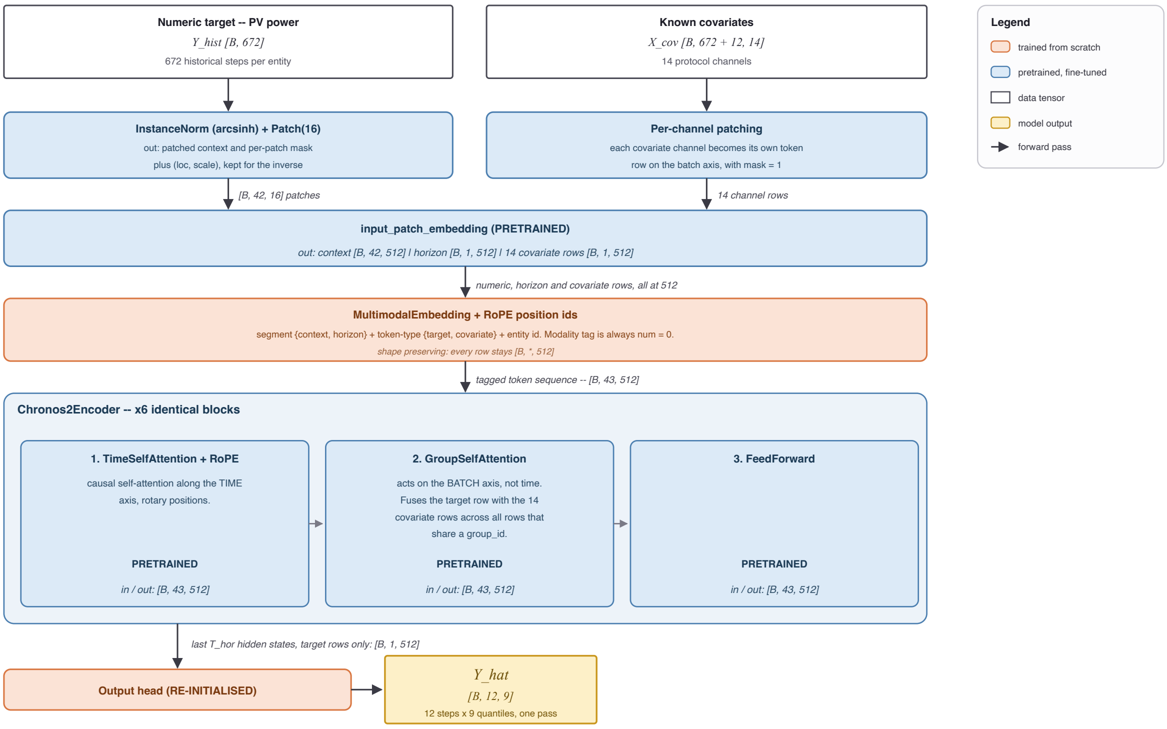

Figure 7: S1, the numeric-only control: the backbone with the visual pathway absent entirely.

S1 (Figure 7) is the same backbone with no imagery at all. It exists to define the baseline against which reliance is measured, separating a multimodal model's advantage from the effect of the fine-tuning recipe. Every subsequent arm is initialised from this checkpoint and trained with the same schedule, so any difference is attributable to the visual pathway.

5.2.2 *S2a — fusion on the batch axis*

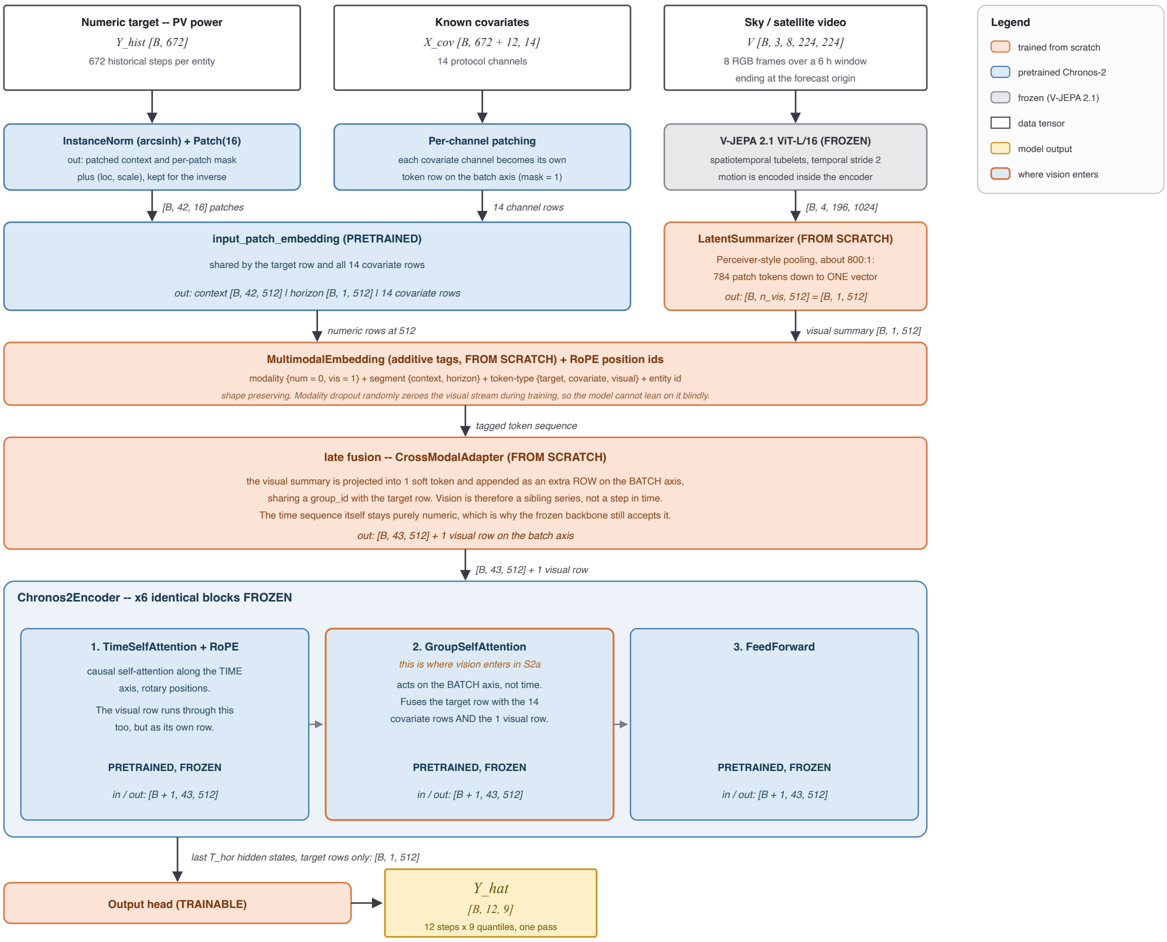

Figure 8: S2a, pooled late fusion: the clip becomes one descriptor presented as an extra parallel channel.

In S2a (Figure 8) the clip is pooled to a single descriptor and presented to the backbone as an extra parallel channel — a summary of the state of the sky, available to every forecast position equally. This is textbook late fusion, structurally equivalent to the pooled pathway of Figure 3, and it was run first precisely **because** it is the standard answer: a null result here is informative rather than an omission.

The rationale for expecting it to work is reasonable. The most obvious use of a satellite view is as a coarse regime indicator — clear, broken, overcast — and a single vector is an adequate carrier for a regime label. The rationale for expecting it to fail is equally clear in hindsight: the pooling operation compresses roughly 800 visual elements into one, and it does so **before anything has been asked of them**. The summary must be computed without knowing which part of the scene is relevant, and by the time the forecaster is in a position to have an opinion, the discarded detail is gone.

5.2.3 *S2b — fusion on the sequence axis*

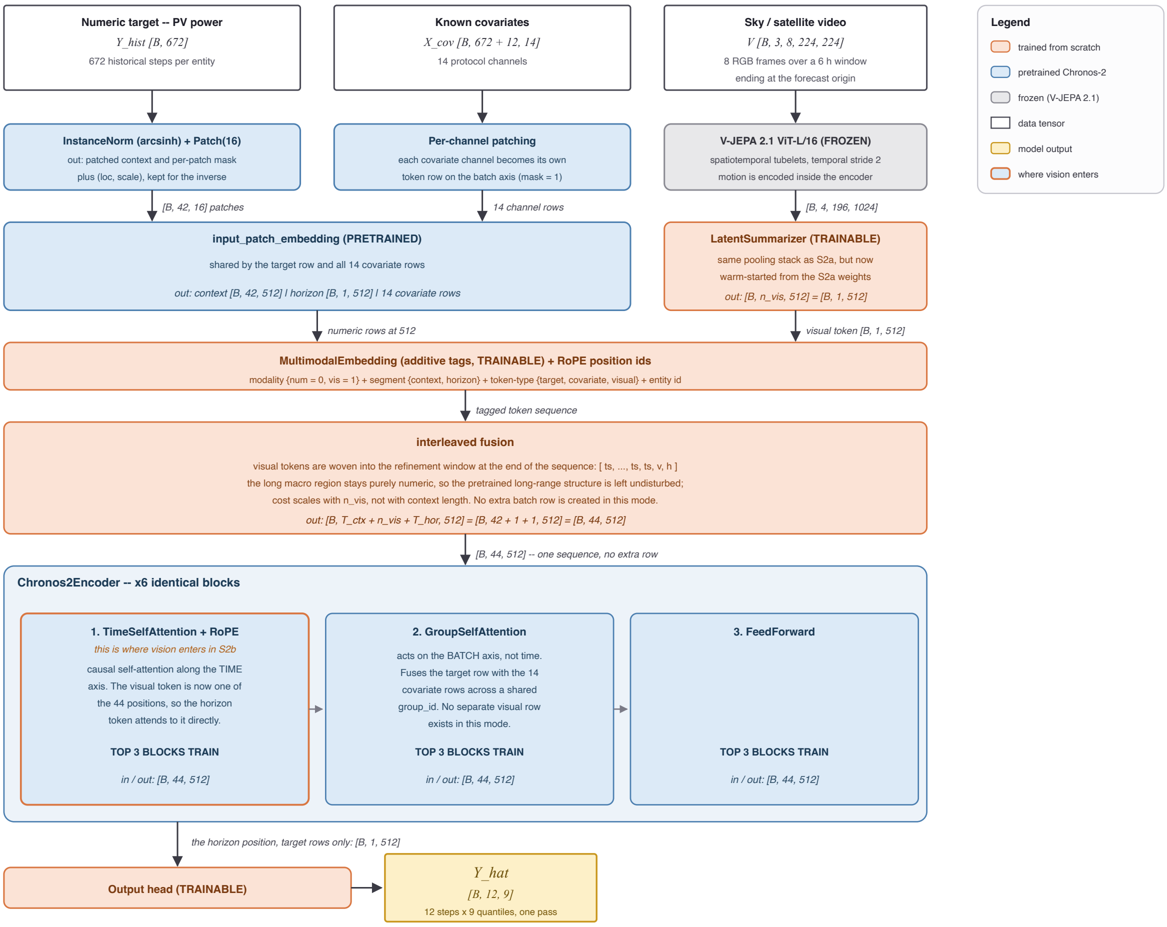

Figure 9: S2b, mid-sequence injection: visual tokens are appended to the token sequence and mixed by ordinary self-attention.

S2b (Figure 9) stops treating vision as a parallel channel and makes it part of the sequence, so that ordinary self-attention can relate visual tokens to numeric ones directly. Two widths were run — one visual token and sixteen — to test whether the failure of S2a was a matter of bandwidth.

Widening the channel from 1 token to 16 did not help; reliance stayed at zero within noise in both settings (Table 6). That comparison rules out the bandwidth explanation and points at something structural: the problem is not **how much** visual information is admitted but **when** it is summarised relative to when it is needed. In both S2a and S2b the imagery is condensed into a fixed representation before the model has formed any view about the forecast, and every forecast position then receives the same representation — the assumption identified in Section 2.6.

5.2.4 *S2c — the forecast queries the sky*

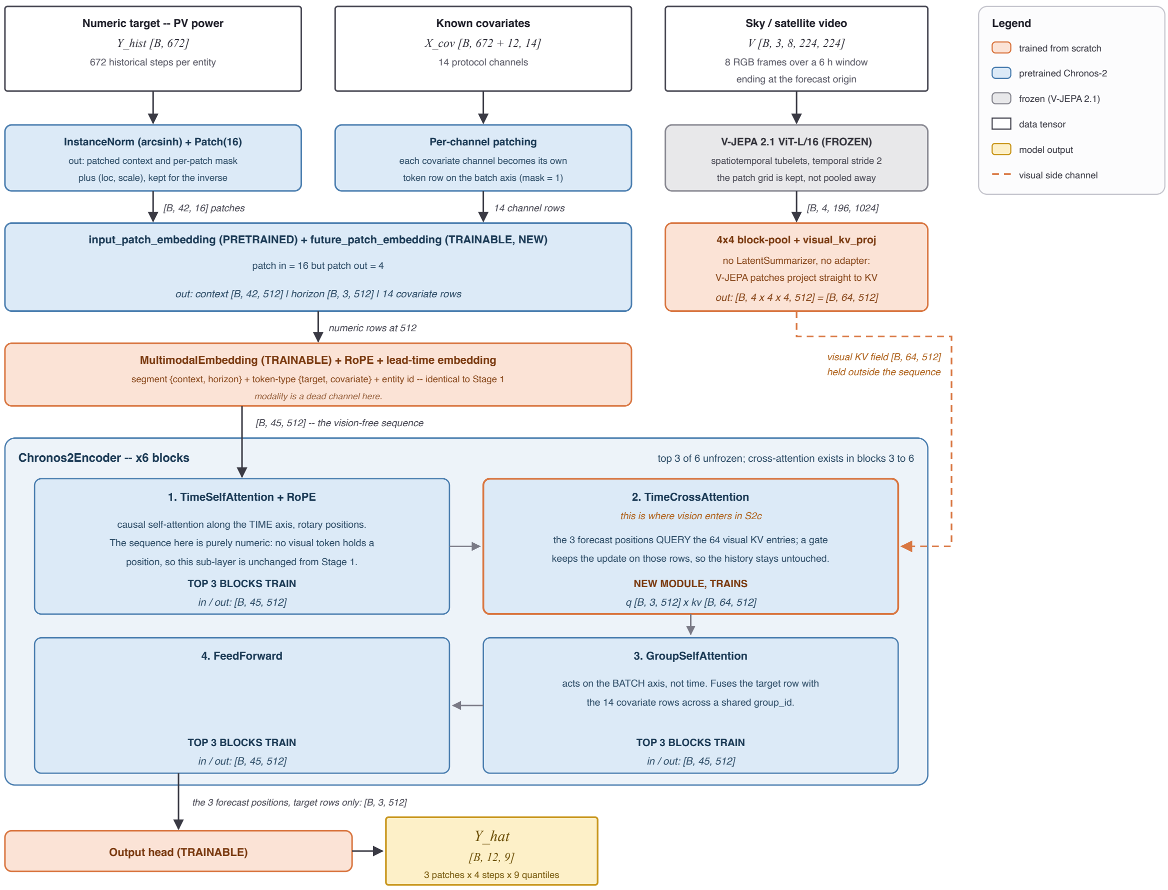

Figure 10: S2c, future-position cross-attention: the imagery is retained unpooled and queried by each lead-time slot.

S2c (Figure 10) changes the direction of the operation. The imagery is not inserted anywhere. It is retained **unpooled** as an external memory — the 14 × 14 patch grid is block-pooled to 4 × 4 and all four temporal slices are kept, giving 64 key–value tokens — and the forecast horizon is subdivided into three lead-time slots. Each slot issues its own cross-attention query against that memory, in each of the backbone's last four encoder blocks. No summarisation happens anywhere: the pooled descriptor and the projection adapter used by the other arms are both bypassed.

What a query is. A query is not made of the quantity being predicted. It is a description of what is being looked for, and it exists before the thing it retrieves. Each forecast slot's query is assembled from two ingredients, both available before any forecast value exists: a learned lead-time identity — a vector attached permanently to the slot at a given horizon, shared across all samples and installations and fixed after training — and the slot's own hidden state after self-attention over the history and the known covariates, which encodes a belief about what is likely to happen rather than the outcome. Figure 11 gives the resulting order of operations.

self-attention populates the slot from history and covariates → slot forms a **query**: lead-time identity + its own belief → query reads the 64 **unpooled visual tokens** → result **gated back** onto that slot alone → projection head emits the number

Figure 11: Order of operations for one lead-time slot in S2c, repeated in each of the last four encoder blocks. Each pass refines the slot's question with what the previous round returned.

Why the slots ask different things. Each lead-time slot receives gradient only from its own horizon's errors. The two-hour slot is penalised for two-hour mistakes, the five-hour slot for five-hour mistakes, and cloud fields relevant at those two horizons sit at different distances from the array. Nothing in the design instructs a slot to attend to cloud edges; the differentiation is the accumulated residue of one horizon's own past errors.

Relation to prior fusion. Conventional late fusion (Figure 1, Figure 3) compresses the auxiliary modality into a shared representation before combining it with the time-series state, which forces one visual summary to serve every horizon even where different horizons need different parts of the scene. Cross-attention alone is not the distinguishing feature either: CrossViViT (Figure 2) already uses it, but its queries originate in the **history** representation and enrich the context before a separate prediction stage. In S2c the queries originate in the **future** positions and read an unpooled memory directly, so each horizon can retrieve different visual evidence.

6 Measuring reliance

After training, the model is frozen and the test set is evaluated twice: once normally, and once with the visual query switched off. The difference between the two passes is the **reliance** — the share of the model's accuracy that depends on it having seen the sky.

This is a stronger instrument than comparing a multimodal model to a unimodal one, because it holds every weight fixed. It cannot be confounded by capacity, initialisation or training schedule; the two passes differ only in whether the imagery was available.

7 Results

7.1 The placement result

Table 6 is the central finding. The reliance column reports the ramp-error improvement attributable to the imagery, measured by the counterfactual pass of Section 6. Reliance is flat across the two arms that summarise the clip in advance, including the sixteen-token variant, and rises by an order of magnitude only when the forecast positions query the memory themselves.

Arm	Where vision enters	Reliance (ramp)
S2a	batch axis, pooled	0.0000 ± 0.0015
S2b	sequence axis, 1 token	0.0006
S2b wide	sequence axis, 16 tokens	0.0002 ± 0.0016
S2c	**future-position queries**	**0.0056 ± 0.0006**

Table 6: The placement ladder. Reliance is the ramp-error improvement attributable to the imagery, measured by switching the visual pathway off at inference on frozen weights. Intervals are across seeds 42–44.

7.2 Ablation status

Table 7 reports the two controls that have been executed. The temporal-shuffle control confirms that the arms whose reliance is zero are genuinely not reading the imagery, and the stale-sky control establishes that S2c's gain is timing-dependent rather than a plant-level constant. Table 8 lists the four further ablations that are configured but not yet launched, together with the outcome each would produce under the hypothesis that the forecast-side query is the operative mechanism.

#	Question it answers	Result
1	Does frame order matter to S2c? (temporal shuffle)	Bit-identical to the unperturbed run at every digit.
2	Is the model reading the sky, or is it reading a plant-level constant?	A stale sky is worse than no sky — vision is read, and read for its timing.

Table 7: Executed ablations and their measured outcomes.

#	Question it answers	Expected behaviour	GPU-h
3	Summariser widened to 4 temporal slices of 4 × 4 blocks — S2c's payload with S2b's fixed-latent queries.	Reliance ≈ 0 ⟶ the mechanism is the forecast-side query.	≈ 24
4	Grid or decoder? S2c with the grid collapsed to 1 × 1	Reliance falls to the S2b level if the grid is what matters; holds if the decoder is.	≈ 24
5	Grid held at 4 × 4, decoder collapsed to one position.	Reliance holds if the grid is the mechanism; falls if the 3-slot decoder is.	≈ 24

#	Question it answers	Expected behaviour	GPU-h
6	Is the grid spatially grounded? Frames swapped with another plant's.	NMAE degrades below the vision-free control and reliance turns negative if the gain is plant-specific; indifference would indicate a generic cloudiness prior.	< 1

Table 8: Open ablations, configured but not launched.

7.3 Placement against the baseline suite

The four arms are scored against 32 baselines under the identical protocol of Table 4. Table 9 reports the multimodal field, Table 10 the unimodal deep and foundation-model field, and Table 11 the tabular, classical and reference tiers. Rank is the global position by skill score across all three tables.

Three observations follow. First, the ordering of the four arms in Table 9 matches the reliance ordering of Table 6 exactly, which is what makes the reliance measure worth trusting as more than an internal diagnostic. Second, the vision-free control alone (SS 0.5230) already outranks every published multimodal forecaster in the suite except Time-VLM — whose second modality is manufactured from the numeric input rather than observed (Figure 4) — which is a statement about the strength of the frozen backbone rather than about those methods. Third, the best ramp NMAE in the entire suite belongs to a unimodal supervised model, iTransformer with forward-shifted covariates (Figure 5), so S2c's advantage is specific to the reliance measurement and does not yet translate into a ramp-metric win; the gap is against an exogenous channel that is aligned to the forecast interval by construction rather than by architecture.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		This work (MMTSFM placement arms)			
1	**MMTSFM S2c** (ours)	satellite	cross-attention from **future** positions to unpooled memory	**0.5470**	0.1461
3	MMTSFM S2b wide (ours)	satellite	sequence axis, 16 appended tokens (self-attention)	0.5352	0.1484
4	MMTSFM S2b (ours)	satellite	sequence axis, 1 appended token (self-attention)	0.5322	0.1487
5	MMTSFM S2a (ours)	satellite	batch axis, pooled vector (group self-attention)	0.5258	0.1487
7	MMTSFM S1 control (ours)	none	— (vision-free control)	0.5230	0.1506
		Multimodal forecasters (satellite & pseudo-image)			
2	Time-VLM [6]	series **rendered as** images	pooled vision + text, query on the temporal side (Figure 4)	0.5404	—
13	Solar-VLM [3]	satellite + text	vision–language fusion, multi-site (Figure 3)	0.4396	0.1514
21	CrossViViT [2]	satellite	cross-attention from history timesteps (Figure 2)	0.3491	—
26	Aurora [21]	several	joint multimodal pretraining	0.2324	—
27	SUNSET [1]	sky/satellite	convolutional precedent; joint encoding (Figure 1)	0.2162	—
28	UniCast [5]	several	prompting a foundation forecaster	0.1211	—
30	VisionTS++ [7]	series **rendered as** images	continual pretraining of a visual backbone	0.0167	—

Table 9: Multimodal leaderboard.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		Supervised deep learning			
6	iTransformer + covariates [8]	covariates	channel-inverted self-attention over variates (Figure 5)	0.5257	**0.1445**
12	PatchTST [22]	none	—	0.4581	0.1543
14	Temporal Fusion Transformer [23]	covariates	gated residual & temporal self-attention	0.4264	0.1605
15	MLP	none	—	0.4219	0.1624
22	DLinear [24]	none	—	0.3231	0.1746
		Retrieval & frozen-backbone adaptation			
9	TS-RAG [9]	retrieved **numeric** history	concatenated to the context	0.4779	—
10	Cross-RAG [10]	retrieved **numeric** history	cross-attention between query and retrievals	0.4768	—
18	CoRA [11]	covariates	residual adapter on frozen backbone	0.3798	0.1624
		Time-series foundation models (zero-shot & fine-tuned)			
8	Chronos-2, fine-tuned [19]	covariates	group self-attention, fine-tuned	0.5042	0.1494
11	Chronos-2, zero-shot [19]	covariates	group self-attention, zero-shot	0.4737	0.1544
20	TTM, fine-tuned [25]	covariates	MLP-Mixer temporal/channel mixing, fine-tuned	0.3601	0.1716
23	TiRex, zero-shot [26]	none	—	0.2873	0.1826
24	TimesFM, zero-shot [27]	none	—	0.2708	0.1902
32	TTM, zero-shot [25]	covariates	MLP-Mixer, zero-shot	−0.0807	0.2922

Table 10: Unimodal deep learning and foundation model baselines.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		Tabular models & classical ML			
16	TabPFN [28]	none	—	0.4063	0.1631
17	LightGBM	covariates	gradient-boosted trees over tabular lags	0.3854	0.1672
19	TabFM ensemble	none	—	0.3626	0.1573
		Reference & statistical baselines			
25	Hourly climatology	none	—	0.2337	0.1665
29	Seasonal naive	none	—	0.1068	0.2056
31	Persistence	none	—	0.0141	0.2550

Table 11: Tabular, classical ML, and reference baselines.

8 Conclusion

Under a frozen-backbone protocol that holds capacity, initialisation and schedule constant, the point at which a visual representation is attached determines whether a forecaster uses it at all. Pooling the clip onto the batch axis and appending it to the token sequence both produce reliance indistinguishable from zero, and the sixteen-token variant shows that this is not a bandwidth limitation. Reliance becomes measurable only when the visual evidence is kept unpooled and queried by the forecast positions themselves. The shared assumption of the prior architectures reproduced in Section 2 — one visual summary, fixed before the horizon is resolved — is therefore a plausible explanation for why frozen multimodal forecasters so often fail to use the modality they were given.

The open ablations of Table 8 are what would separate the two candidate mechanisms inside S2c, the unpooled spatial grid and the per-lead-time decoder, and they are the immediate next step.

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